

LIMIT-SKETCHABLE INFINITY CATEGORIES

David Martínez-Carpena

Carles Casacuberta Javier J. Gutiérrez

International Category Theory Conference 2024

Santiago de Compostela



UNIVERSITAT DE
BARCELONA

Limit sketches

Limit sketches $\longleftrightarrow$ Essentially algebraic theories

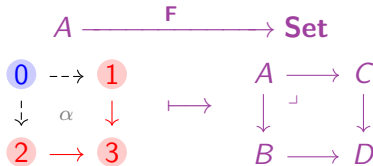
A **limit sketch** (Bastiani and Ehresmann 1972) is a pair $(A, \mathcal{L})$ of a small category A and a set of cones $\mathcal{L}$ over A .

A **model** of a limit sketch is a functor $F : A \rightarrow \mathbf{Set}$ which sends cones of $\mathcal{L}$ to limits of **Set**.

Example. Let A be the small category generated by the square α :

Observe that α is a cone
with **apex** and **diagram**

Then $(A, \{\alpha\})$ is a sketch 🍷



A model F of the sketch $(A, \{\alpha\})$ is a pullback of sets 🍷

Presentability

A category C is **accessible** if it is locally small and there exists a regular cardinal κ such that:

- ✓ C admits κ -*filtered* colimits.
- ✓ There is a small full subcategory of κ -*compact objects* that generates C under κ -*filtered* colimits.

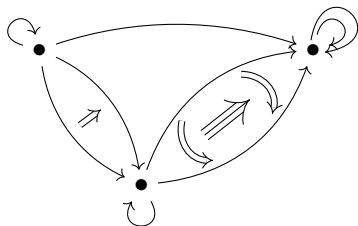
A category is **(locally) presentable** if it is accessible and cocomplete.

Presentable	Accessible but not presentable	Not accessible
Set	Fields	Top
Grp	Hilbert spaces	
sSet		
RMod		

Higher categories

Informally, an ∞ -**category** (or $(\infty, 1)$ -**category**) has objects and:

- ✓ n -morphisms between $(n - 1)$ -morphisms for all $n \geq 1$,
- ✓ Composition, identities and associativity of n -morphisms weakly up to a $(n + 1)$ -morphism for all $n \geq 1$.
- ✓ n -morphisms are invertible up to higher morphisms $\forall n > 1$.



Representation theorem

We say that a category is **limit-sketchable** if it is equivalent to the category of models of some limit sketch.

Theorem

A category is presentable if and only if it is limit-sketchable.

- › Originally proved by Gabriel and Ulmer 1971.
- › Revisited by Adamek and Rosicky 1994.

Goal

Generalize this result to ∞ -categories

Plan

1. Limit ∞ -sketches
2. Presentable ∞ -categories
3. Representation theorem

Plan

1. Limit ∞ -sketches
2. Presentable ∞ -categories
3. Representation theorem

Limits and colimits

Let $\mathcal{C}$ be an ∞ -category, and $\mathcal{I}$ be a small ∞ -category. Then:

- › The **constant diagram** $\underline{x} : \mathcal{I} \rightarrow \mathcal{C}$ over an object $x \in \mathcal{C}$ sends everything in $\mathcal{I}$ to x and its higher identities.
- › A **cone** of a diagram $D : \mathcal{I} \rightarrow \mathcal{C}$ and apex $x \in \mathcal{C}$ is a natural transformation $\alpha : \underline{x} \Rightarrow D$.
- › A **limit cone** of $D : \mathcal{I} \rightarrow \mathcal{C}$ is a cone $\ell : \underline{\lim D} \Rightarrow D$ such that, for any $x \in \mathcal{C}$, ℓ induces an equivalence

$$\mathrm{Hom}_{\mathcal{C}}(x, \lim D) \xrightarrow{\sim} \mathrm{Nat}_{\mathrm{Fun}(\mathcal{I}, \mathcal{C})}(\underline{x}, D).$$

Cocones and **colimit cocones** are defined as cones and limit cones in the opposite ∞ -category.

Limit ∞ -sketches

A **limit ∞ -sketch** (Joyal 2008) $\Sigma = (\mathcal{A}, \mathcal{L})$ is a small ∞ -category $\mathcal{A}$ equip with a set $\mathcal{L}$ of cones.

Let $\mathcal{C}$ be a $\mathcal{L}$ -complete ∞ -category. A **model** of a limit ∞ -sketch $\Sigma = (\mathcal{A}, \mathcal{L})$ in $\mathcal{C}$ is a functor $F : \mathcal{A} \rightarrow \mathcal{C}$ which sends each cone in $\mathcal{L}$ to a limit cone in $\mathcal{C}$.

$\text{Mod}(\Sigma, \mathcal{C}) := \infty\text{-category of models of } \Sigma \text{ in } \mathcal{C}$

$\text{Mod}(\Sigma) := \infty\text{-category of models of } \Sigma \text{ in } \mathbf{Spaces}$

We say that an ∞ -category is **limit ∞ -sketchable** if it is equivalent to the ∞ -category of models of some limit ∞ -sketch.

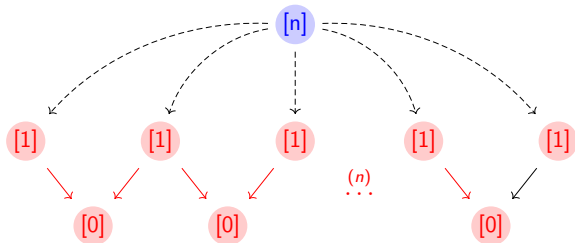
An ∞ -sketch $(\mathcal{A}, \mathcal{L})$ is **normal** if all the cones of $\mathcal{L}$ are limit cones.

Examples

- **Algebraic theories** (Rosicky 2007): Monoid objects (A_∞ -spaces), commutative monoid objects (E_∞ -spaces), group objects (∞ -groups), R-modules, ...
- **Algebraic patterns** (Chu and Haugseng 2021): Category objects (Segal spaces), operad objects (Segal dendroidal spaces), ...
- Spectrum object (Spectra)
- Univalent category objects (Complete Segal spaces) and univalent operad objects (Complete Segal dendroidal spaces)
- ∞ -sheaves $\simeq$ hypercomplete ∞ -topos

Examples: Category objects

Let $\mathcal{C}$ be a complete ∞ -category, and α_n be the cone over Δ^{op} with **apex** and **diagram** for all $n \in \mathbb{N}$:



Then $\Sigma = (\Delta^{\text{op}}, \{\alpha_n \mid n \in \mathbb{N}\})$ is a limit ∞ -sketch, and a model $F : \Delta^{\text{op}} \rightarrow \mathcal{C}$ is a simplicial object in $\mathcal{C}$ such that

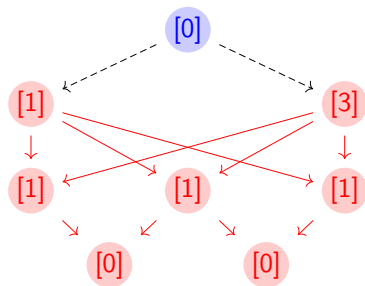
$$F_n \xrightarrow{\sim} F_1 \times_{F_0} F_1 \times_{F_0} \cdots \times_{F_0} F_1. \quad (\text{Segal condition})$$

$\text{Mod}(\Sigma, \mathcal{C}) \simeq$ **Category objects on $\mathcal{C}$**

$\text{Mod}(\Sigma) \simeq$ **Segal spaces**

Examples: Univalent category objects

Let $\mathcal{L}_S$ be the set of cones of the previous example, and β be the cone over Δ^{op} with **apex** and **diagram**:



Then $\Sigma = (\Delta^{\text{op}}, \mathcal{L}_S \cup \{\beta\})$ is a limit ∞ -sketch, and a model $F : \mathcal{A} \rightarrow \mathcal{C}$ is an internal category in $\mathcal{C}$ such that

$$\begin{array}{ccc}
 F_0 & \xrightarrow{\quad} & F_3 \\
 \downarrow & \lrcorner & \downarrow \\
 F_1 & \longrightarrow & F_1 \times_{F_0}^{d_1, d_1} F_1 \times_{F_0}^{d_0, d_0} F_1
 \end{array}$$

$\text{Mod}(\Sigma, \mathcal{C}) \simeq$ **Univalent category objects on $\mathcal{C}$**

$\text{Mod}(\Sigma) \simeq$ **Complete Segal spaces**

Plan

1. Limit ∞ -sketches
2. Presentable ∞ -categories
3. Representation theorem

Filtered colimits and compact objects

Let κ denote a regular cardinal and $\mathcal{C}$ an ∞ -category.

- A space $X \in \mathbf{Spaces}$ is κ -**small** if $\pi_0(X)$ and $\pi_n(X, x)$ are κ -small sets for all $n > 0$ and $x \in X$.
- $\mathcal{C}$ is κ -**small** if the hom-spaces $\mathrm{Hom}_{\mathcal{C}}(x, y)$ are κ -small for all $x, y \in \mathcal{C}$, and the set of iso classes of objects of $\mathcal{C}$ is κ -small.
- $\mathcal{C}$ is κ -**filtered** if, for every κ -small ∞ -category $\mathcal{I}$, every diagram $D : \mathcal{I} \rightarrow \mathcal{C}$ admits a cocone $\alpha : D \Rightarrow \underline{x}$.
- $\mathcal{C}$ admits κ -**filtered colimits** if it admits $\mathcal{K}$ -indexed colimits, for every κ -filtered ∞ -category $\mathcal{K}$.
- An object $x \in \mathcal{C}$ is called κ -**compact** if the hom-space functor $\mathrm{Hom}_{\mathcal{C}}(x, -) : \mathcal{C} \rightarrow \mathbf{Spaces}$ preserves κ -filtered colimits.

Accessibility

Definition (1)

An ∞ -category $\mathcal{C}$ is **accessible** if it is locally small and there is a regular cardinal κ such that:

- ✓ $\mathcal{C}$ admits κ -filtered colimits.
- ✓ There is some essentially small sub- ∞ -category of κ -compact objects which generates $\mathcal{C}$ under κ -filtered colimits.

Definition (2)

An ∞ -category $\mathcal{C}$ is **accessible** if it is equivalent to the free cocompletion of $\mathcal{A}$ under κ -filtered colimits $\mathrm{Ind}_{\kappa}(\mathcal{A})$ for some regular cardinal κ and some small ∞ -category $\mathcal{A}$.

Theorem (Joyal 2008, Lurie 2009). (1) $\iff$ (2)

Presentability

Definition

An ∞ -category is **presentable** if it is accessible and cocomplete.

Example

- The nerve of any presentable 1-category is presentable.
- The ∞ -category of homotopy types **Spaces** is presentable.
- If $\mathcal{A}$ is a small ∞ -category and $\mathcal{C}$ is a presentable ∞ -category, then $\mathrm{Fun}(\mathcal{A}, \mathcal{C})$ is presentable.
- Any ∞ -topos is presentable.

Plan

1. Limit ∞ -sketches
2. Presentable ∞ -categories
3. Representation theorem

Representation theorem

Theorem

- (a) *For each limit ∞ -sketch Σ , $\text{Mod}(\Sigma)$ is presentable.*
- (b) *For any presentable ∞ -category $\mathcal{C}$, there is a normal limit ∞ -sketch Σ such that $\mathcal{C} \simeq \text{Mod}(\Sigma)$.*

Corollary

- (a) *An ∞ -category is presentable if and only if it is limit ∞ -sketchable.*
- (b) *If $\mathcal{C}$ is presentable and Σ is a limit ∞ -sketch, then $\text{Mod}(\Sigma, \mathcal{C})$ is presentable.*
- (c) *For each limit ∞ -sketch Σ , there exists a normal limit ∞ -sketch Θ such that $\text{Mod}(\Sigma) \simeq \text{Mod}(\Theta)$.*

Work in progress

- 💡 **Generalization:** A sketch is a limit sketch with a set of cocones which are sent to colimit cocones by any model. There is the following result due to Lair (revisited later by Makkai-Paré and Adámek-Rosický):

$$\text{Accessible} \iff \text{Sketchable}$$

Can this result be generalized to the ∞ -categorical setting?

- 💡 Formalize this work with a proof assistant which supports synthetic ∞ -categories like `rzk`.

References

-  Gabriel, Peter and Friedrich Ulmer (1971). *Lokal präsentierbare Kategorien*. Vol. 221. Lecture Notes in Mathematics.
-  Adamek, Jiří and Jiří Rosicky (1994). *Locally Presentable and Accessible Categories*. Vol. 189. Lecture Notes in Mathematics.
-  Joyal, André (2008). *The Theory of Quasi-Categories and its Applications*. Barcelona: Lectures at CRM.
-  Rosicky, Jiří (2007). “On homotopy varieties”. *Advances in Mathematics* 214.2.
-  Chu, Hongyi and Rune Haugseng (2021). “Homotopy-coherent algebra via Segal conditions”. *Advances in Mathematics* 385.
-  Lurie, Jacob (2009). *Higher Topos Theory*. Annals of Mathematics Studies 170.

Thank you for listening!

This work is supported by the *Departament de Recerca i Universitats de la Generalitat de Catalunya* (2021 SGR 00697) and from the *Agencia Estatal de Investigación* (AEI) under grant PID2020-117971GB-C22.

LIMIT-SKETCHABLE INFINITY CATEGORIES

David Martínez-Carpena

Carles Casacuberta Javier J. Gutiérrez

International Category Theory Conference 2024

Santiago de Compostela



UNIVERSITAT_{DE}
BARCELONA